

Data Preparation Techniques

Outliers, Binning, Transforming, Encoding, Scaling & Shuffling

AWS Machine Learning Associate • Part 10

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What We'll Cover



Handling outliers



Binning



Transforming



Encoding & one-hot encoding



Scaling



Shuffling

Handling Outliers: Spotting Them

A data point that sits far from the rest. It may be a genuine rare event, or a mistake like a typo or sensor glitch.

Z-Score Method

$$|z| > 3$$

Flag values more than about 3 standard deviations away from the mean.

IQR Method

Below Q1 − 1.5×IQR
or above Q3 + 1.5×IQR

Flag anything outside this range. Box plots show it visually.

Handling Outliers: Your Options



Remove

Makes sense when they are clearly errors



Cap (Winsorize)

Replace extreme values with a limit, like the 95th percentile



Transform

A log transform pulls extreme values closer in



Keep

Sometimes the outlier is the signal, like a fraudulent transaction

Never delete outliers blindly. Ask: is this an error, or real information?

Binning

Group continuous values (like exact ages) into a few buckets:

Under 18 | **18 – 35** | **36 – 60** | **Over 60**

Equal-Width Binning

Every bin covers the same range of values, like 0–20, 20–40, 40–60.

Quantile Binning

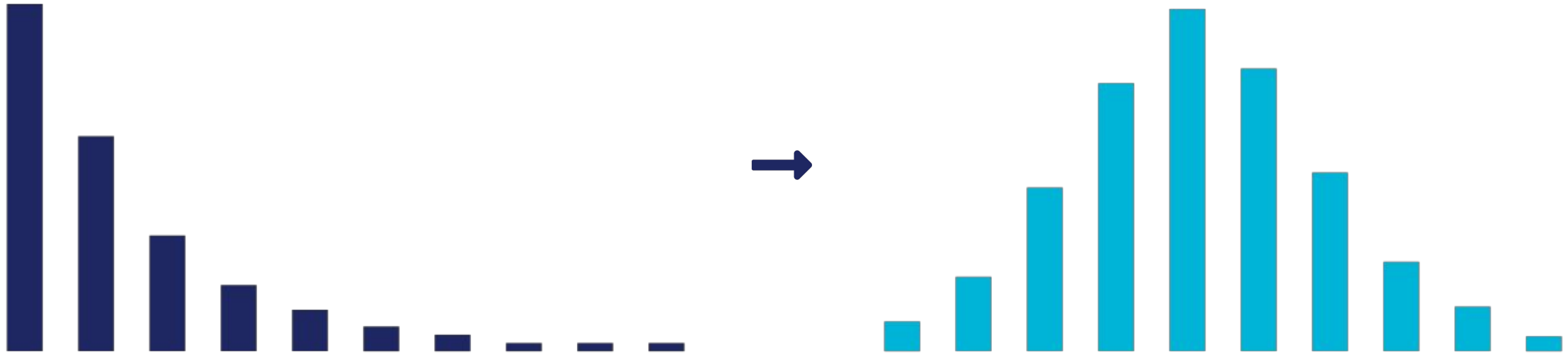
Every bin holds roughly the same number of data points.

Benefits: less noise, softer outliers, captures non-linear patterns. Trade-off: some detail is lost.

Transforming

Skewed (before)

After a log transform



Log
for heavy right skew

Square root
a milder option

Box-Cox
finds a good power for you

Helps models that expect roughly normal data, and reduces the pull of outliers.

Encoding

Most algorithms cannot work with text labels, so categories must become numbers.

Label / Ordinal Encoding

One integer per category. Best when there is a real order.

```
Small  → 0
Medium → 1
Large  → 2
```



The Catch

On unordered categories like colors or cities, the model may wrongly assume one is "bigger" than another.

That is where one-hot encoding comes in.

One-Hot Encoding

Color	Red	Blue	Green
Red	1	0	0
Blue	0	1	0
Green	0	0	1
Blue	0	1	0

Each row has exactly one column switched on.

- **No order implied**
  Treats every category as equal
- **Best for**
  Colors, cities, product types
- **Watch out**
  Hundreds of categories means hundreds of columns

Scaling

Normalization (Min-Max)

```
(x - min) / (max - min)
```

Squeezes values into a 0 to 1 range.

Standardization (Z-Score)

```
(x - mean) / std
```

Result has a mean of 0 and a standard deviation of 1.

Needed: KNN, SVM, neural networks, gradient descent

Usually not needed: tree-based models

Tip: fit the scaler on training data only, then apply it to the test data to avoid leakage.

Shuffling

Before: sorted by class



After: shuffled



Fair splits

Train and validation sets both see every class

No order learning

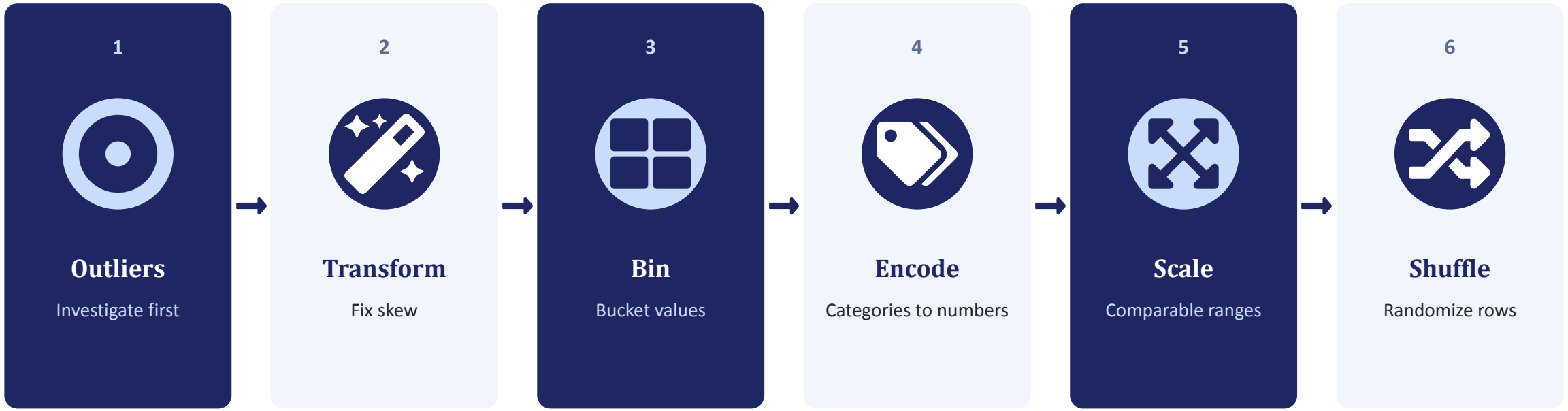
The model learns patterns, not row positions

Exception: time series

Order carries meaning, so do not shuffle

For neural networks, reshuffle at the start of every epoch.

Putting It Together



A solid default order. Adjust it for your data and your model.

Recap

- Outliers need a decision, not an automatic delete.
- Binning, transforming & encoding reshape features for the model.
- Scaling & shuffling keep training fair and stable.

Thanks for watching!